Feature Importance Ensemble

Sam Tritto

OMSA Practicum – Mid Term Report

Fall 2023

Outline

- Objective
- Data Sources
- Pre-Processing
- EDA
- Ensemble Methods
- Outputs
- Current Status & Future Work

Objective

Each year our business gathers a handful of metrics which help our business leaders make informed decisions throughout the year. We often consider these our north star metrics as they align well to our business principles. Since these metrics are only gathered yearly, it can be difficult to use them for business decisions throughout the year. Instead, I will work toward understanding which measurable metrics that we currently monitor at the monthly level correlate strongest toward the north star metrics to serve as proxy metrics.

I will do this by creating a feature importance ensemble and comparing the results from several different algorithms. Feature importance itself is a niche category of feature selection, where we do not wish to remove any features, rather understand the importance of each feature toward the target metric. It will be important for us to use an ensemble of different methods to pick up on different trends and relationships between the metrics; our data is commonly complex and non-Normal. Some methods in the feature importance ensemble may include correlations, various regressions (lasso, quantile), tree-based models (Random Forest, XGBoost, LightGBM), cross validation, stepwise elimination, forward selection, backward selection, regularization, permutation importance, shapely values, etc…

By the end of the analysis, we will have a handful of measurable metrics that truly matter and correlate strongest toward our north star metrics. These metrics will be used as proxies throughout the year by various business units in their projects.

Data

Data Sources, Pre-Processing, and EDA

Data Sources

Internal

- 293 Monthly Business Metrics
  - From different business units
  - Some are included because of an existing hypothesis from leaders and teams, so we aim to validate
  - Put together in SQL by managers and analysts (not me, but I was part of the discussions)
- 1998 Locations
- Monthly for the Current Fiscal Year
- 23,760 Records in Total
- 8 Target Metrics
  - Run separately
  - More can be added for ad hoc analysis throughout the year

External

- 13 Metrics
  - Many more are pulled for future use
- For Context and as Control Variables
- Matched to our Locations
  - Through their geographies
  - Used Google BigQuery's public geo data for proximities, latitudes/longitudes, and boundaries by zip-code, county, state, etc…
- Through Downloads & APIs
  - ETL Process into many clean and useable tables
  - Automated for weekly data pulls (python, BigQuery, selenium, etc…)
- Governmental & Institutional Sources
  - NLRB
  - BLS
  - Cornell
  - FBI
  - HUD

Pre-Processing

Categorical Data

- Dummy Variables
- Label Encoding (for ordinal data)

Feature Engineering

- Normalize count metrics by headcount per location
- Combine multiple metrics into new metrics (rates)

Outliers

- Conditional Ensemble
  - Based on the skew of the distribution of each metric
  - z-score, modified z-score, and IQR
- Only remove a handful of locations with the most number of outlier columns
- Maybe keep all outliers after investigation

Imputation

- Zero Imputation
  - For count metrics and division errors resulting from headcount normalization
- Expanding Geo Median Imputation
  - For geographical data points
  - Starting at the smallest geography, then expanding outward
  - By median average
- MICE or KNN Imputation
  - Used for the remaining metrics
  - MICE is regression based with 4 iterations
  - KNN with 10 nearest neighbors

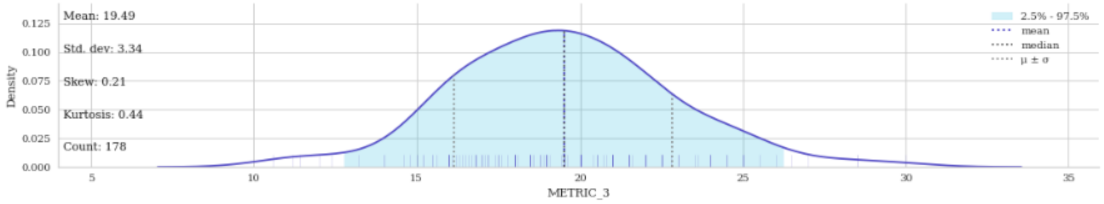
General Cleaning

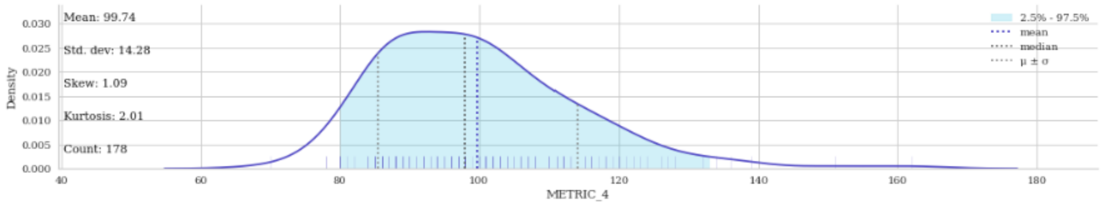
- Data Types
- Duplicates

EDA

Distribution Plots

- Visualize distribution of each metric
- Looking for symmetry, normality, skew, and outliers



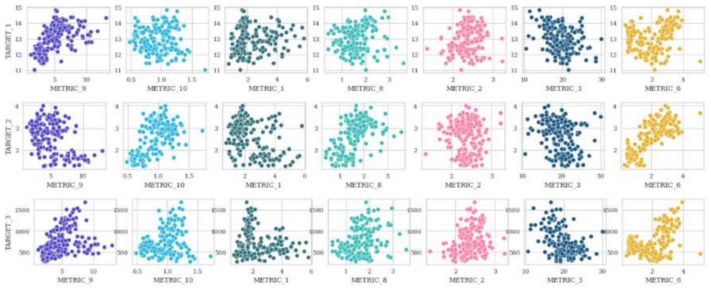


Descriptive Stats

- Count, mean, std, min, max, and percentiles of each metric
- Looking for oddities (values should be positive but there's a negative, etc…)

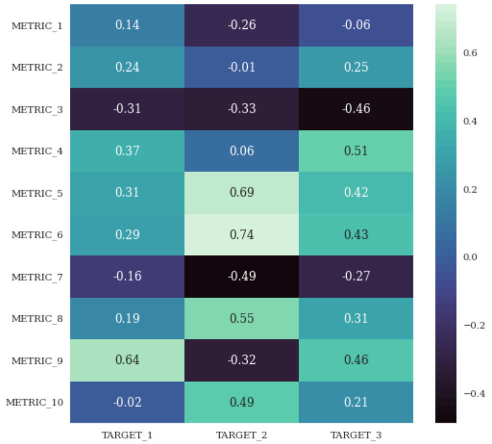
Scatter Plots

- Relationship between features and targets
- Looking for linearity and outliers



Correlations

- Between all features to understand multicollinearity
- Between targets and features as an estimate of importance



NOTE: Synthetic data has been used for all vizualizations & outputs

Modeling

Ensemble Methods and Outputs

Ensemble Methods

Correlation Based

1. Spearman Correlation

Decision Tree Based

1. Permutation Importance
2. BorutaSHAP

Regression Based

1. Backward Quantile Regression
2. Forward Quantile Regression
3. LassoCV

Final Model for Inference
(Not Selection)

1. Stepwise Quantile Regression

Ensemble Methods

Correlation Based

- Independently calculated so multicollinearity is not an issue

Decision Tree Based

- Can choose between XGBoost, Random Forest, and LightGBM
- Overfitting can be an issue
- Perform Random Grid Search to tune parameters
- Evaluate through performance metrics
  - RMSE
  - MAE

Regression Based

- Multicollinearity is an issue
- Use VIF with 5.0 threshold to remove highly correlated variables
- Assign these removed variables the same importance as their highest respective correlated variable using a threshold
- Evaluate through R^2 and Residual Analysis

Ensemble Methods

Correlation

Spearman Correlation

There's no linearity assumption

Does a good job at approximating relationship quickly

Values between -1 and 1

Arbitrary thresholds are used for importance cutoff (> 0.1 or < -0.1)

Forward Selection

Quantile Regression

Based on median

Avoids some assumptions of OLS

Heavily transformed all variables with Box-Cox and Standardization

Iteratively adds one metric to the regression

Keeps metric with smallest p-value if p-value is < 0.05

Backward Selection

Quantile Regression

Based on median

Avoids some assumptions of OLS

Heavily transformed all variables with Box-Cox and Standardization

Iteratively removes one metric from the regression

Removes metric with largest p-value if p-value is < 0.05

Stepwise Selection

Quantile Regression

Based on median

Avoids some assumptions of OLS

Just standardization to preserve coefficients for inference

Iteratively adds and removes metrics from the regression

Keeps metric if p-value is < 0.05, drops if > 0.05

LassoCV

OLS Regression with penalty

Heavily transformed all variables with Box-Cox and Standardization

Uses cross validation to find optimal alpha value for regularization

All non-important metrics coefficients have been forced to 0 and thus removed

Permutation Importance

Decision tree

Metrics are shuffled out one by one

A random feature is created and added to the model each iteration, used as threshold

Importance is the average decrease in model score for each metric when removed

Good for non-linear data at different scales

BorutaSHAP

Decision tree

Random features are created from existing metrics iteratively and done multiple times

Each metric is evaluated against these shadow features through a hypothesis test (binomial distribution)

SHAP values are the metrics used to evaluate against the shadow features and assign importance

Ensemble Methods

Things I've Tried But Didn't Work Out

PCA	Generalized Additive Model (GAM)	Ordinary Least Squares (OLS)	AIC/BIC Selection	Decision Tree Feature Importance
Has a linearity assumption Ambiguous feature importance thresholds, and hard to determine the number of PCs to accept To capture much of the varaince would bring in too many metrics (80% explained varaince brings in 120 metrics)	Great to perform regressions and avoid assumptions of OLS Used Optuna to find the optimal number of knots per metric (298 * 2 = 596 parameters to tune, that's too many!) Was hoping to use this model for inference, and the goal was to average the splines. After inspection of the results this didn't seem like a feasible approach.	Many assumptions we not met Interpretable if assumptions are met Tried heavy transformations and still could not pass a residual analysis check Would've been great since we can perform stepwise/forward/backward selection with AIC/BIC	Since I ended up using Quantile Regression (statsmodels) there was no AIC/BIC available to use for selection, since there is no likelihood function because the noise added to the model is not Gaussian This resulted in using p-value for selection which is not ideal, but we decided it would be best given the shape of our data and the fact that we can capture possibly missed metrics through the tree-based models in the ensemble	Each decision tree model used has it's own feature importance output, however these have all been show to be problematic for highly correlated variables One correlated variable would show as important but it's counter part would show as unimportant Even a model like Random Forest suffers from this issue

Ensemble Methods

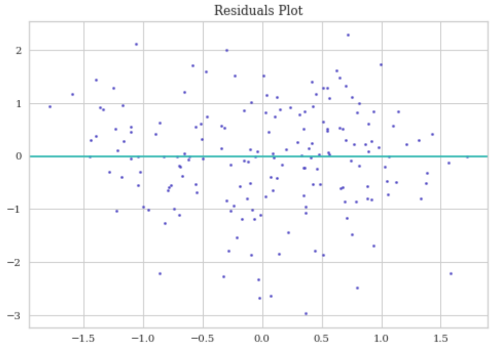
R^2

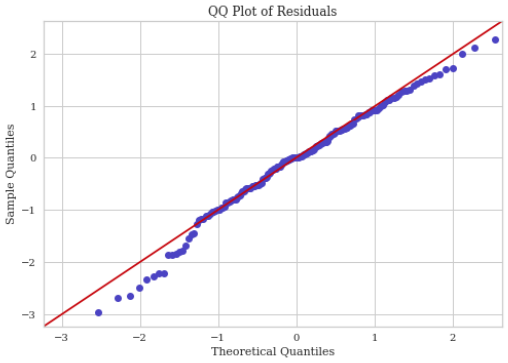
QuantReg Regression Results

Dep. Variable:	TARGET_1	**Pseudo R-squared:**	0.3690
Model:	QuantReg	**Bandwidth:**	0.5696
Method:	Least Squares	**Sparsity:**	1.767
Date:	Wed, 27 Sep 2023	**No. Observations:**	178
Time:	15:49:52	**Df Residuals:**	171
		Df Model:	6

Selected Feature's Importance

Residual Analysis





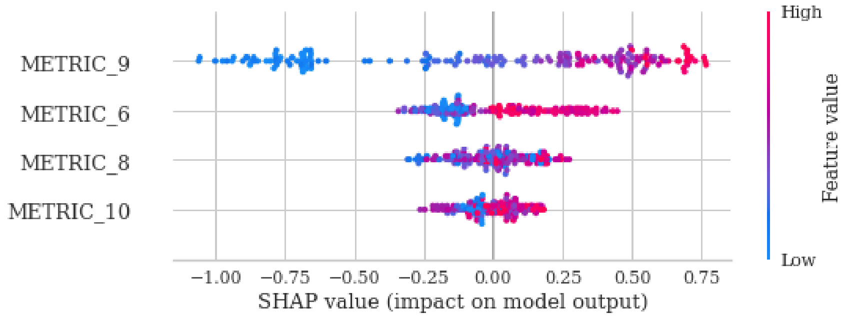

```
NORMALITY TESTS ON RESIDUALS

SHAPIRO-WILK TEST       statistic=0.986, p-value=0.072
Sample looks Normal with alpha = 0.05

D'AGOSTINO's K2 TEST    statistic=5.213, p-value=0.074
Sample looks Normal with alpha = 0.05

ANDERSON-DARLING TEST   statistic: 0.611
Sample looks Normal with alpha = 0.1
```

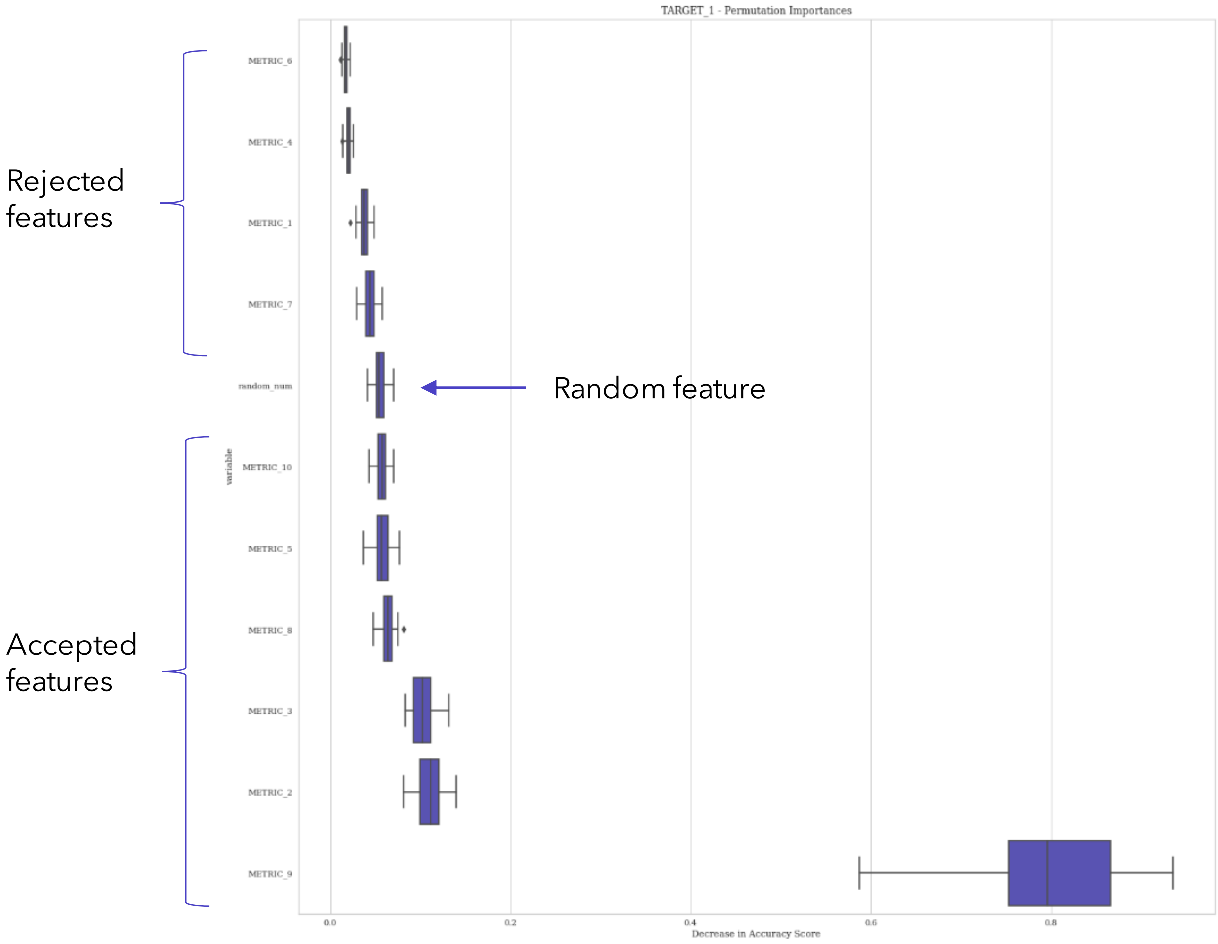
SHAP



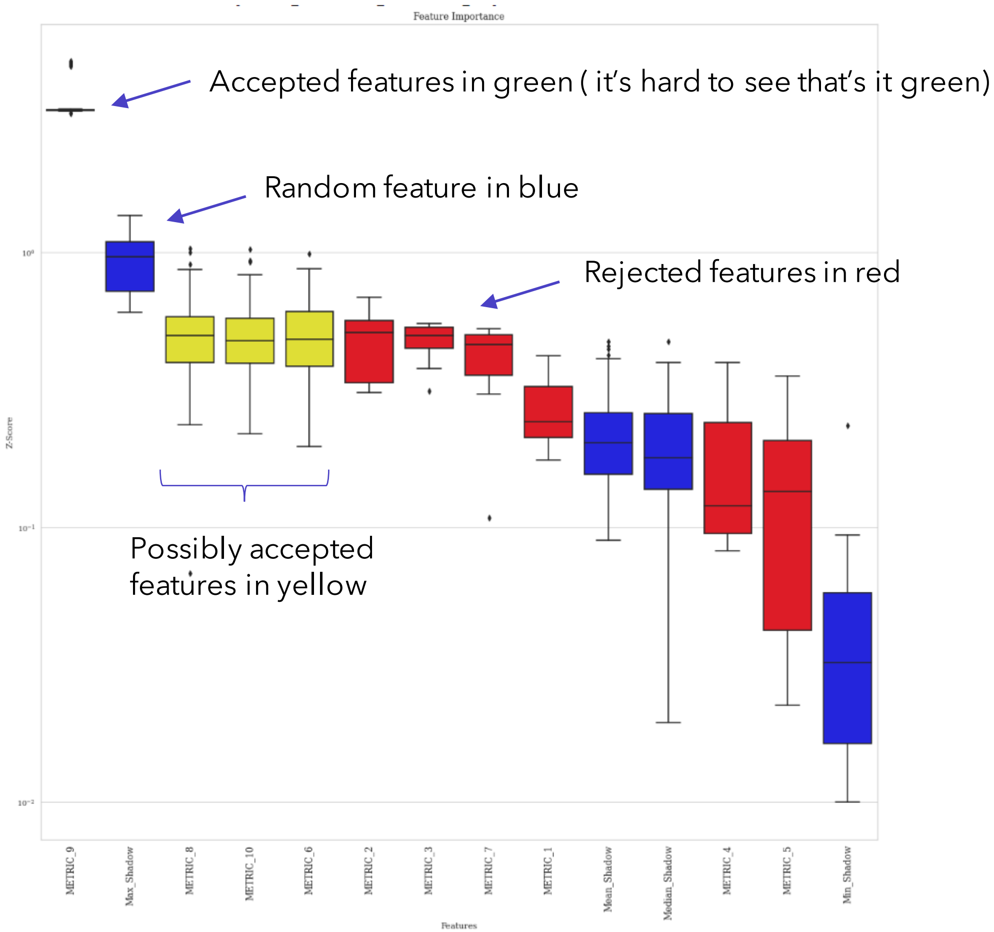
NOTE: Synthetic data has been used for all vizualizations

Ensemble Methods

Permutation Importance



BorutaSHAP



NOTE: Synthetic data has been used for all vizualizations & outputs

Outputs

Ensemble Model Importance Votes

- Count the discrete votes for each metric per model

Final Regression

- For inference
- We'd like to know the estimated effect of each metric toward the target metrics – on average… holding all else constant

Ensemble Model Importance Score

- Scale all feature importance scores between 0.1 and 1.1
- Add them and assign the score

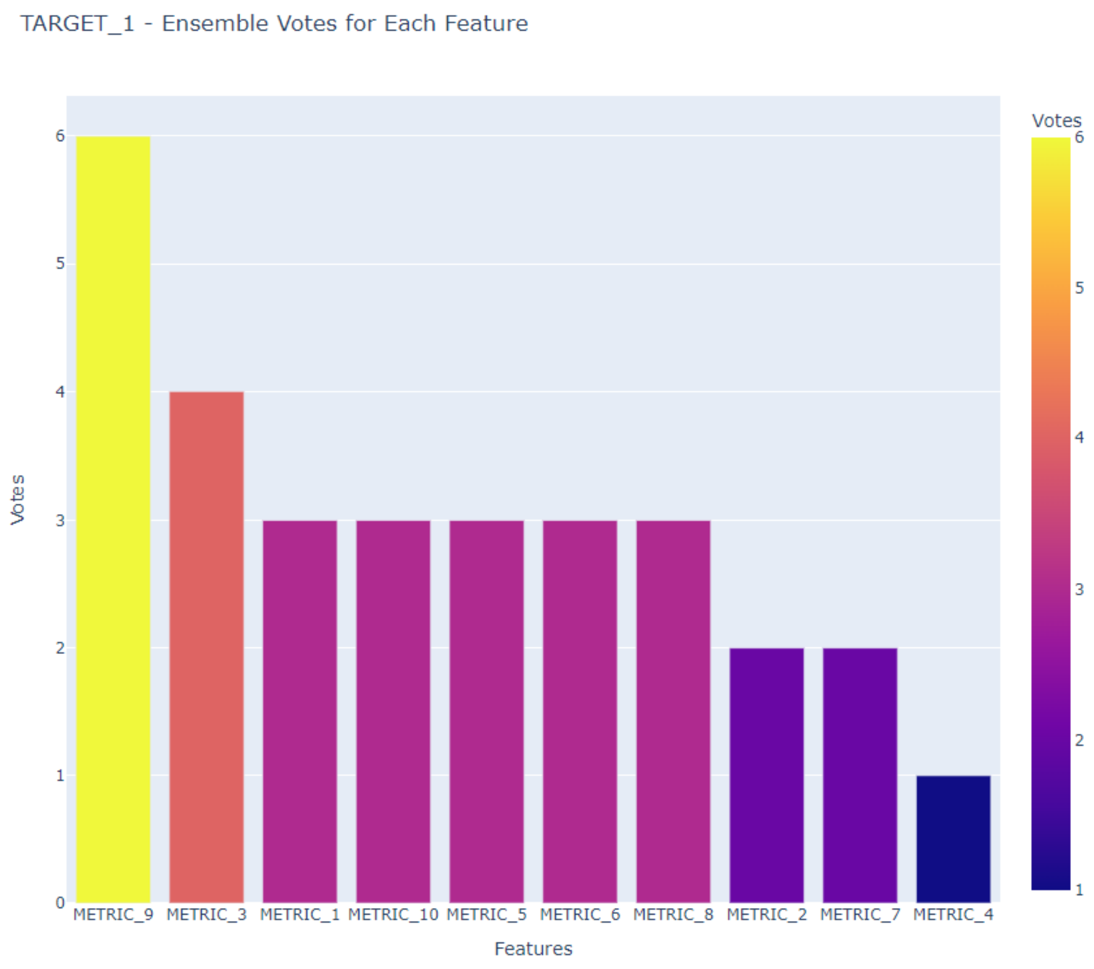
Prescriptive Analytics

- We'd like to be able to identify thresholds and benchmarks for each metric towards the target metrics

Outputs

Ensemble Model Importance Votes

- Count the votes for each metric per model
- Discrete
- Can use a minimum vote as a threshold for accepting features into the final list and final regression model
- From 293 metrics to 25 metrics that matter

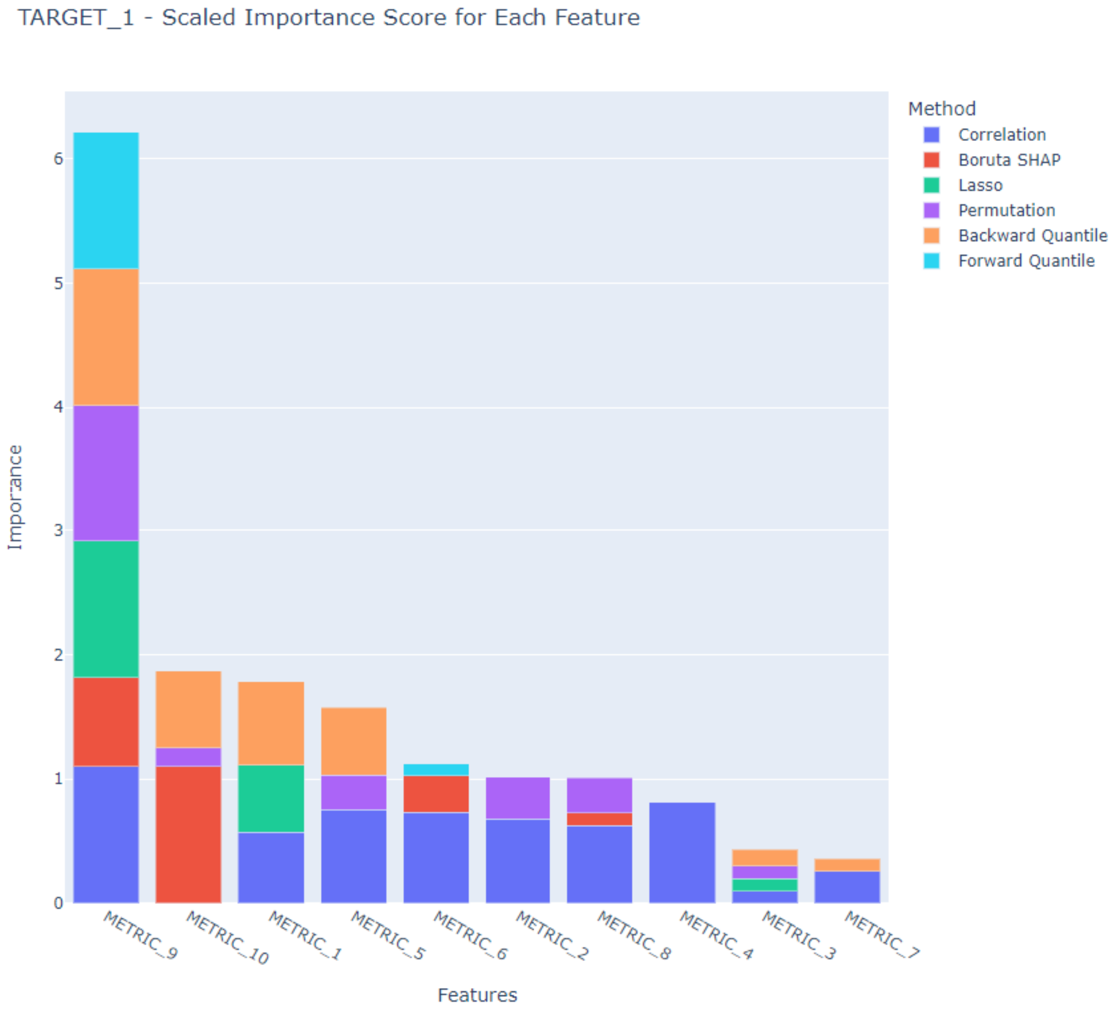


NOTE: Synthetic data has been used for all vizualizations & outputs

Outputs

Ensemble Model Importance Score

- Scale all feature importance scores between 0.1 and 1.1
- Continuous
- Add them and assign the score
- Can use a minimum score as a threshold for accepting features into the final list and final regression model



NOTE: Synthetic data has been used for all vizualizations & outputs

Outputs

Final Regression Inference

- Only use the features chosen from the votes of the feature importance ensemble
  - At least 2 models voted for the feature
  - About 25 metrics in total chosen
- Bring in categorical data points as factors to increase R^2
- Stepwise Selection
- Final summary() is used for inference
- We can now understand the estimated effect of each selected metric on our target metrics

QuantReg Regression Results

Dep. Variable:	TARGET_1	**Pseudo R-squared:**	0.3140
Model:	QuantReg	**Bandwidth:**	0.5436
Method:	Least Squares	**Sparsity:**	1.592
Date:	Wed, 27 Sep 2023	**No. Observations:**	178
Time:	16:26:48	**Df Residuals:**	174
		Df Model:	3

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	13.0289	0.060	218.424	0.000	12.911	13.147
METRIC_9	0.4450	0.062	7.161	0.000	0.322	0.568
METRIC_3	-0.3855	0.067	-5.757	0.000	-0.518	-0.253
METRIC_2	0.2734	0.069	3.944	0.000	0.137	0.410

Estimated Average Effect:

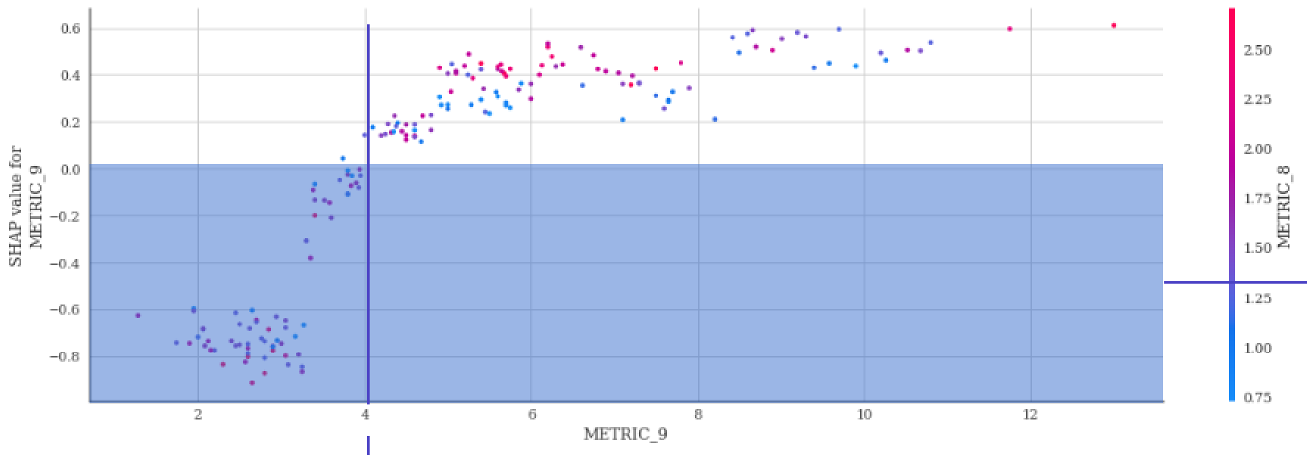
"As METRIC_2 increases by one unit, on average we can reasonably expect our TARGET_1 metric to increase by 0.2734 units holding all other factors constant."

NOTE: Synthetic data has been used for all vizualizations & outputs

Outputs

Prescriptive Analytics

- SHAP Values vs Metric Values
  - Used to create new metrics (above or below benchmark)



The threshold where the metric starts to associate with positive contributions toward the prediction on the target variable.

- LIME Values vs Metric Values
  - Coming Soon

The shaded area contains negative SHAP Values and represent negative additive contributions toward the prediciton from our model towards the target metric

Sample Prescriptive Benchmark:
"We should aim for metric values above 4"

NOTE: Synthetic data has been used for all vizualizations & outputs

Current Status & Future Work

Current Status

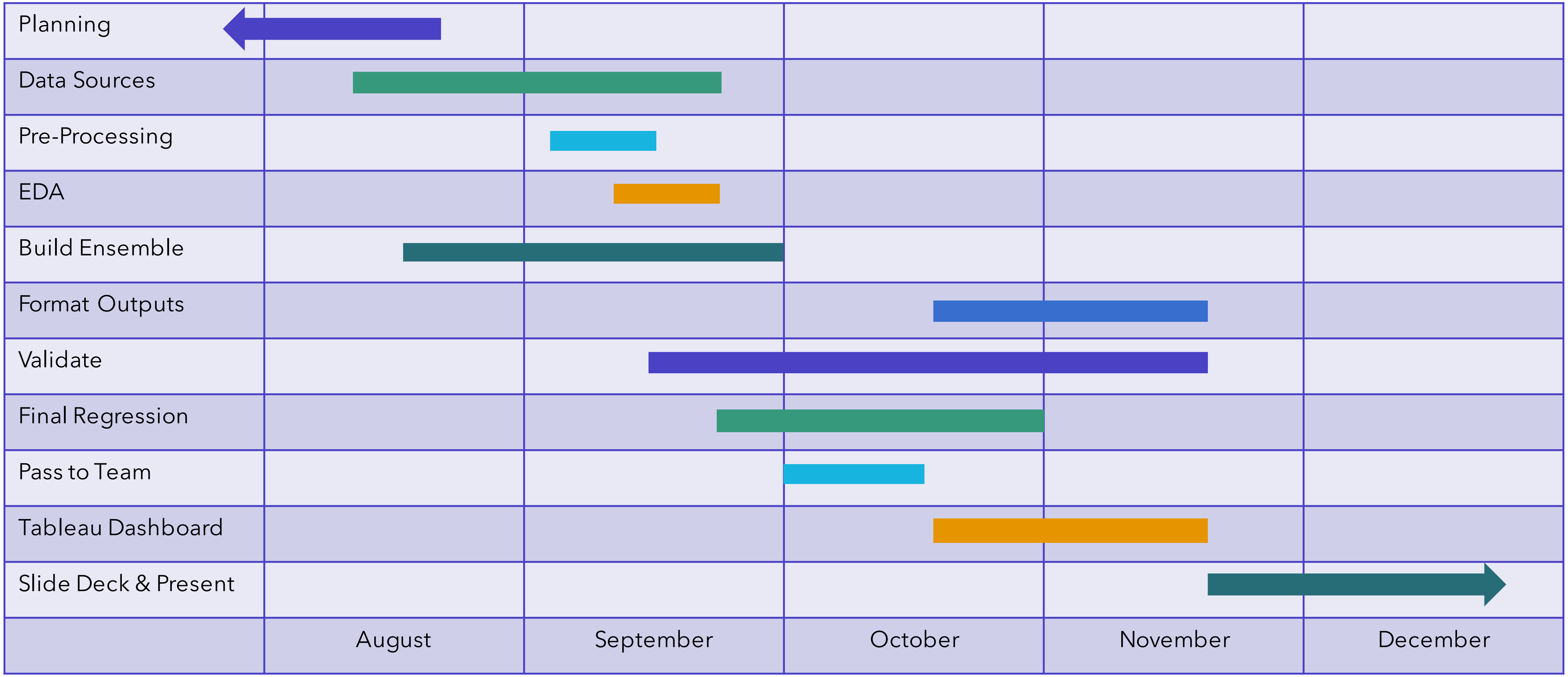
- Finalizing the template
- Working through minor bugs and validating
- Working on final regression to improve R^2
- Documentation

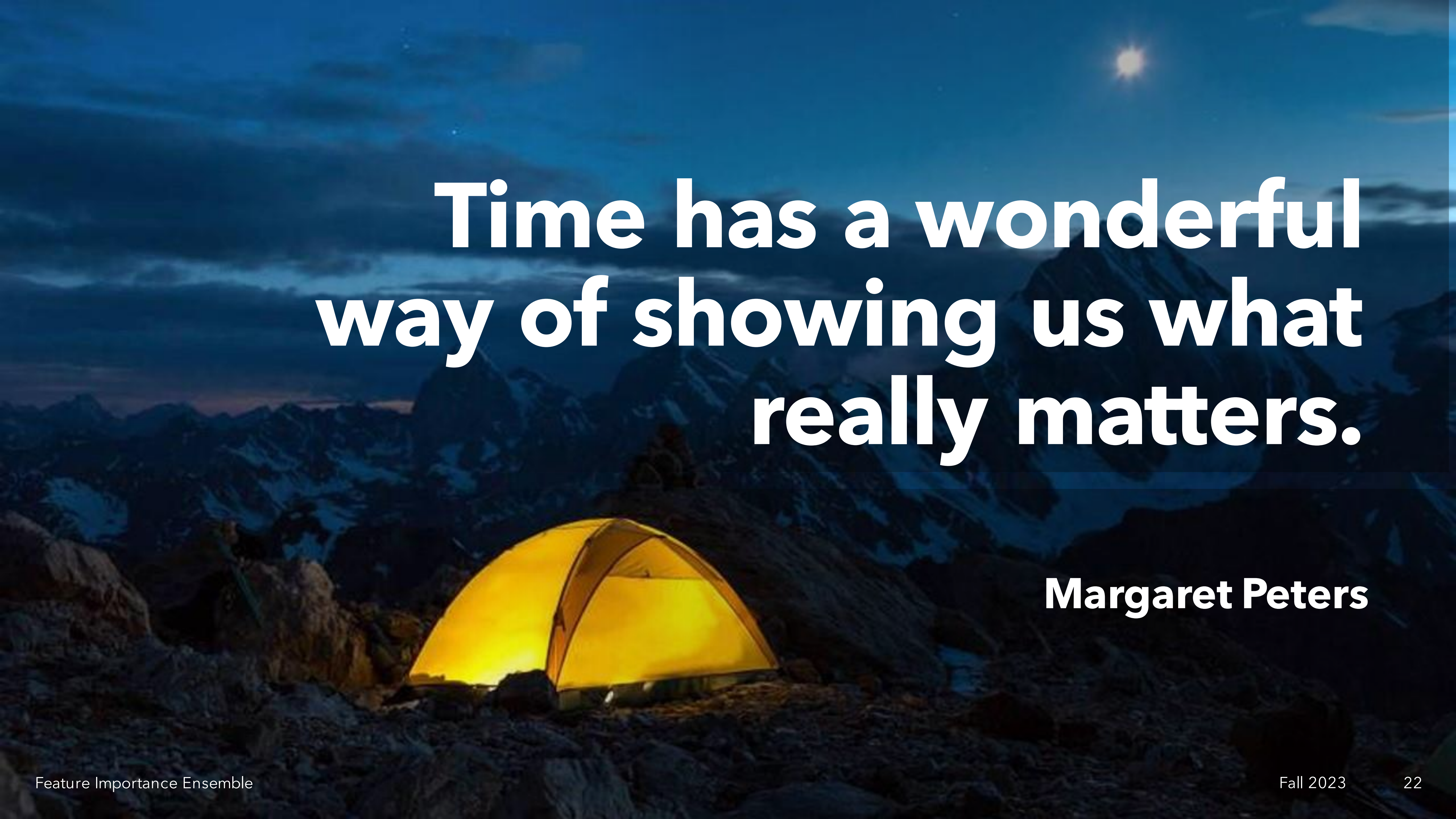
Future Work

- Pass onto the team so they can adapt to their business units
- Take feedback from the team and update the template
- Develop report and slide deck based on findings
- Create benchmark metrics based off thresholds in prescriptive analysis
- Final written report

Current Status & Future Work

Gantt Chart





Time has a wonderful way of showing us what really matters.

Margaret Peters

Summary

Overall, I’m happy with where the project is at now and the progress made thus far. I’ve started with 293 metrics and boiled them down to about 25 that truly matter. I feel that everything is in place for me to be able to deliver insights to our leaders which can shape their decision making for the better and throughout the year. Having a clear plan as to what our outputs will be and how we can best communicate these findings moving forward has helped tremendously. I’ve really enjoyed being able to dive deep into the methodologies and choose the appropriate models for our unique data. Putting it all together so that the notebook has a nice flow has been the most challenging aspect so far. I’m really looking forward to fine tuning the final regression model, finding smart ways to communicate the inference points from the model to our business leaders, and telling an interesting story.

Sam Tritto

stritto3@gatech.edu

Thank you

Appendix

Literature Review

Not a formal review, but rather a list of links that I've used to build my knowledge on. I'll work on something more in-depth for the final report.

- https://arxiv.org/pdf/2207.07038.pdf
- https://www.jstatsoft.org/article/view/v036i11/
- https://towardsdatascience.com/boruta-shap-an-amazing-tool-for-feature-selection-every-data-scientist-should-know-33a5f01285c0
- https://stats.stackexchange.com/questions/94130/does-boruta-feature-selection-in-r-take-into-account-the-correlation-between-v
- https://pyod.readthedocs.io/en/latest/pyod.models.html#module-pyod.models.ecod
- https://arxiv.org/pdf/2201.00382.pdf
- https://scikit-learn.org/stable/auto_examples/inspection/plot_permutation_importance_multicollinear.html
- https://towardsdatascience.com/boruta-explained-the-way-i-wish-someone-explained-it-to-me-4489d70e154a
- https://github.com/Ekeany/Boruta-Shap
- https://towardsdatascience.com/best-practice-to-calculate-and-interpret-model-feature-importance-14f0e11ee660
- https://towardsdatascience.com/seaborn-pairplot-enhance-your-data-understanding-with-a-single-plot-bf2f44524b22
- https://towardsdatascience.com/assumptions-in-ols-regression-why-do-they-matter-9501c800787d
- https://statmodeling.stat.columbia.edu/2009/07/11/when_to_standar/
- https://towardsdatascience.com/using-shap-values-to-explain-how-your-machine-learning-model-works-732b3f40e137
- https://medium.com/dataman-in-ai/explain-your-model-with-the-shap-values-bc36aac4de3d
- https://datascience.stackexchange.com/questions/63024/permutation-feature-importance-vs-randomforest-feature-importance
- https://stats.stackexchange.com/questions/550499/how-shap-values-behave-in-terms-of-multicollinearity-in-trees-ensemble-gradien
- https://klib.readthedocs.io/en/latest/modules.html
- https://stats.stackexchange.com/questions/155028/how-to-systematically-remove-collinear-variables-pandas-columns-in-python
- https://betterdatascience.com/feature-importance-python/
- https://www.researchgate.net/post/What_does_negative_factor_loading_imply_in_PCA
- https://inria.github.io/scikit-learn-mooc/python_scripts/linear_regression_non_linear_link.html
- https://jekel.me/piecewise_linear_fit_py/examples.html#non-linear-standard-errors-and-p-values
- https://pypi.org/project/pwlf/
- https://jekel.me/piecewise_linear_fit_py/examples.html#non-linear-standard-errors-and-p-values
- https://stackoverflow.com/questions/54147278/how-to-explain-variables-weight-from-a-linear-discriminant-analysishttps://stackoverflow.com/questions/54147278/how-to-explain-variables-weight-from-a-linear-discriminant-analysis
- https://towardsdatascience.com/squeezing-more-out-of-lime-with-python-28f46f74ca8e
- https://arxiv.org/pdf/2006.05714.pdf
- https://medium.com/analytics-vidhya/lasso-regression-fundamentals-and-modeling-in-python-ad8251a636cd
- https://www.statology.org/lasso-regression-in-python/
- https://www.scikit-yb.org/en/latest/api/regressor/residuals.html

Appendix

Literature Review

Part 2

- https://stats.stackexchange.com/questions/303261/aic-bic-for-quantile-regression
- https://www.statsmodels.org/dev/generated/statsmodels.regression.quantile_regression.QuantReg.html#statsmodels.regression.quantile_regression.QuantReg-attributes
- https://stackoverflow.com/questions/3701170/stepwise-regression-using-p-values-to-drop-variables-with-nonsignificant-p-value
- https://datascience.stackexchange.com/questions/24405/how-to-do-stepwise-regression-using-sklearn/24447#24447?newreg=73e38e7b7b694b7fb595e3f81404da79
- https://pypi.org/project/miceforest/
- https://python.plainenglish.io/automated-feature-selection-for-machine-learning-in-python-2ad4bcfac19a
- https://en.wikipedia.org/wiki/Generalized_additive_model
- https://crunchingthedata.com/when-to-use-generalized-additive-models/#:~:text=Generalized%20additive%20models%20are%20generally,for%20nonparametric%20machine%20learning%20models.
- https://stats.stackexchange.com/questions/380426/when-to-use-a-gam-vs-glm
- https://www.statsmodels.org/stable/gam.html
- https://multithreaded.stitchfix.com/blog/2015/07/30/gam/